

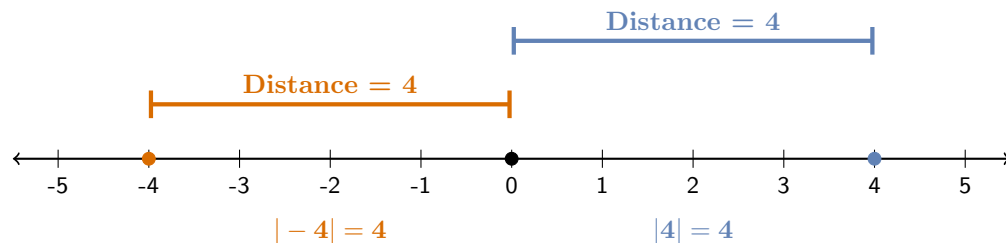
Absolute Value and Opposites- This sucks, nothing works!

Absolute Value

If you're cutting a piece of wood to be 10 inches long, being 0.5 inches too long is just as problematic as being 0.5 inches too short. In both cases, the **size** of the error is 0.5 and this is what matters. The concept of **absolute value** can be used to represent the **magnitude** of a mistake, regardless of whether it was an overshoot or undershot.

The **absolute value of a number** is its **distance from zero** on a number line. Since distance is always positive or 0, so is absolute value. Absolute value is represented by two vertical bars and can be interpreted as the **size, strength** or **magnitude** of a number.

The number line below shows the absolute values of -4 and 4 . Since their absolute values are the same, the numbers -4 and 4 have the same strength or magnitude.



Alt text: A horizontal number line from negative 5 to 5 with solid dots plotted at negative 4, 0, and 4. Above the line, an orange bar spans from negative 4 to 0 and a blue bar spans from 0 to 4, both labeled "Distance = 4". Below the line, it states that the absolute value of negative 4 is 4 and the absolute value of 4 is 4.

To find the absolute value of a number, think of the number line as a path and each integer as a step. The absolute value of a number is how many steps you would have to take to travel from the number to zero. For example, you would have to take 2 steps to travel from -2 to 0, so the absolute value of -2 is 2.

$$| - 2 | = 2$$

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Problem 1 Find the absolute value of each number.

$$|-1| = 1$$

$$|7| = 7$$

$$|-55| = 55$$

Problem 2 You are building a custom bookshelf and need to cut boards that are 36 inches long. A board will be usable if its length is off by no more than 0.25 inch of the target. Suppose you cut the first board 35.8 inches long. The difference between the actual board length and the target length is the error.

$$\begin{aligned}\mathbf{Error} &= 35.8 - 36 \\ &= -0.2 \text{ inches}\end{aligned}$$

The negative sign indicates that you cut the board too short. However, it only matters if the board is useable or not, so we take the **absolute value** to find the **distance** from the target length. This distance is called the **absolute error**.

$$\begin{aligned}\mathbf{Absolute Error} &= |35.8 - 36| \\ &= |-0.2| \\ &= 0.2 \text{ inches}\end{aligned}$$

Is this board usable? **Hint:** A board is usable if the absolute error is less than or equal to 0.25 inches.

Multiple Choice:

- (a) Yes, the absolute error is less than 0.25 inches. ✓
- (b) No, the absolute error is greater than 0.25 inches.

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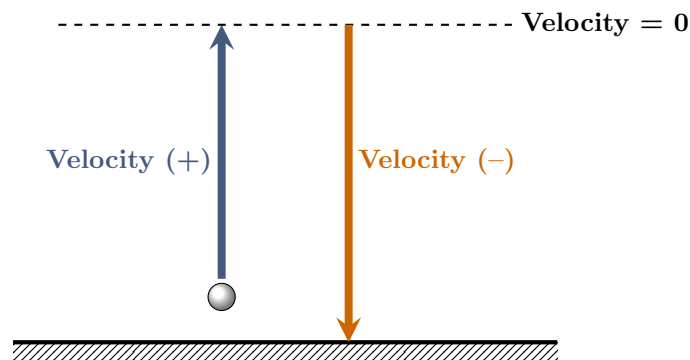
Extra Insight: The Relationship Between Speed and Velocity

In physics, the motion of an object is often described by its **velocity**, which indicates its speed *and* direction. When an object moves **upward** or to the **right**, its velocity is **positive** and when it moves **downward** or to the **left**, its velocity is **negative**. We use absolute value to turn velocity into speed.

$$\text{Speed} = -\text{Velocity}-$$

Speed is the magnitude or size of velocity.

When a ball is thrown straight up in the air, its velocity is positive as it moves upward, zero when it reaches its highest point and negative when it falls back to Earth. If, after 6 seconds, the velocity of the ball is -15 feet per second, we know that it is falling back to Earth, since its velocity is negative. Since $|-15| = 15$ the ball is traveling at a speed of 15 feet per second 6 seconds after it was thrown upward.



Alt text: A physics diagram illustrating vertical projectile motion relative to a ground line. A ball is above the ground on the left. A thick blue upward arrow represents the ball's ascending path and is labeled "Velocity" with a plus sign next to it. A dashed horizontal line marks the peak of the flight, labeled "Velocity = 0". A thick orange downward arrow on the right represents the descending path and is labeled "Velocity" with a minus sign next to it.

Problem 3 A model rocket was launched straight up in the air. After 8 seconds, the velocity of the rocket was recorded as -12 feet per second. In which direction was the rocket moving 8 seconds after it was launched?

Multiple Choice:

- (a) Downward ✓

Absolute Value and Opposites- This sucks, nothing works!

(b) *Upward*

What was the rocket's speed at that moment?

Speed = 12 feet per second

Opposites

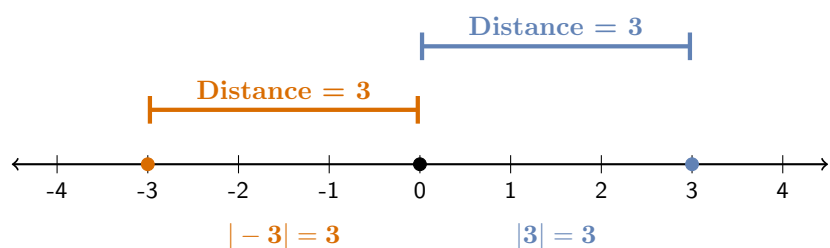
Definition 1 (Opposites). Two numbers that are the same distance from zero but on opposite sides of zero are called **opposites**.

The opposite of a number is found by changing its sign.

The phrase “**the opposite of**” is written mathematically as a dash ($-$).

Example 1. State two numbers that are opposites and explain why they are opposites.

Explanation. The numbers 3 and -3 are opposites. They lie the same distance from zero on the number line, but in **opposite** directions, and they have the same absolute values.



Alt text:

A horizontal number line from negative 4 to 4 with solid dots plotted at negative 3, 0, and 3. Above the line, an orange bar spans from negative 3 to 0 and a blue bar spans from 0 to 3, both labeled “Distance = 3”. Below the line, it states that the absolute value of negative 3 is 3 and the absolute value of 3 is 3.

Properties of Opposites

- (a) Opposites have the same absolute value because they are equidistant from zero.
- (b) One is positive and the other is negative.
- (c) Adding a number and its opposite always results in zero: $a + (-a) = 0$.
- (d) Zero is unique because it is its own opposite: $0 = -0$.

Absolute Value and Opposites- This sucks, nothing works!

Problem 4 Complete the table.

Number	Opposite	Absolute Value
-15	15	15
8	-8	8
0	0	0
-2.5	2.5	2.5

Definition 2 (The Three Meanings of a Dash $-$). In mathematics, the **dash** is used in three different ways depending on its role.

- (a) **Negative:** When a dash is attached directly to a number, it describes its location on the number line and is **read** as **negative**.
- (b) **Opposite:** A dash placed before parentheses, absolute value or a variable represents an action: changing the sign or multiplying by -1 and is **read** as **the opposite of**.
- (c) **Subtraction:** A dash placed between two numbers represents the operation of subtraction and is **read** as **minus**.

Remark 1. The expression $-x$ is sometimes read as negative x , rather than the opposite of x . However, keep in mind that the expression can be positive, negative or even 0, depending on the value of x .

Problem 5 Identify the correct way to read each expression.

- (a) $18 - 5$ is read as (Negative 5/ Eighteen minus five ✓/ The opposite of 5).
- (b) -20 is read as (Negative 20 ✓/ The opposite of negative 20/ Minus 20).
- (c) $-|-7|$ is read as (The opposite of the absolute value of negative 7 ✓/
The opposite of the absolute value of 7. / The absolute value of negative 7).
- (d) $-k$ is read as (Dash k / Minus k / The opposite of k ✓)

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Definition 3 (The Double Negative Property). For any number a ,

$$-(-a) = a$$

The expression $-(-a)$ is read as “the opposite of negative a .”

Problem 6 Use the Double Negative Property to simplify the following expressions.

$$-(-14) = 14$$

$$-(-100) = 100$$

$$-(-0) = 0$$

Problem 7 How is the expression $-(-22)$ read?

Multiple Choice:

- (a) Negative negative 22
- (b) Minus negative 22
- (c) The opposite of negative 22 ✓
- (d) The opposite of 22

Problem 8 If $x = -10$, what is the value of $-(-x)$?

Multiple Choice:

- (a) 10
- (b) -10 ✓
- (c) 0

Absolute Value and Opposites- This sucks, nothing works!

Feedback(attempt): Remember: $-(-x) = x$, therefore since x started as -10 , the final result is also -10 .

Warning 1. A common mistake is to treat absolute value like parentheses. Keep in mind that **parentheses** are used for **grouping**, whereas **absolute value** is an **operation**. Make certain you understand the difference between the following cases.

Double Negative with Parentheses

Since we have parentheses, we simply find the opposite of negative 7, which is 7.

$$-(-7) = 7$$

Double Negative with Absolute Value

To find the opposite of the absolute value of negative 7, start with the absolute value: $|-7| = 7$, then find the opposite of that result. The opposite of 7 is -7 therefore,

$$-|-7| = -7.$$

Absolute Value vs. Opposite

While absolute value always results in a nonnegative (zero or positive) number, taking the opposite of a number changes its sign. Therefore, an opposite can be positive, negative or 0. Note that for the number 0, both its absolute value and its opposite are 0. Later, you will see how these two concepts are used together to add integers.

Below, the concept of an opposite has been used to write a rule for finding absolute value. Notice that this rule doesn't rely on distance.

Procedure 1 (Finding the Absolute Value of a Number). Positive Numbers and 0:

The absolute value of a positive number or 0 is the number itself.

Negative Numbers:

The absolute value of a negative number is the **opposite** of that number.

Problem 9 True or False If a number is negative, its opposite must be positive.

Multiple Choice:

(a) True ✓

(b) False

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Feedback(correct): Exactly! The opposite flips the sign, so a negative becomes positive.

Problem 10 True or False Two different numbers can have the same absolute value.

Multiple Choice:

(a) True ✓

(b) False

Feedback(correct): Yep! The numbers 2 and -2 have the same absolute values. In fact, all opposites have the same absolute values.

Problem 11 Which has a greater value: the opposite of -10 or the absolute value of -10 ?

Multiple Choice:

(a) Neither, they are the same. ✓

(b) The opposite of -10 .

(c) The absolute value of -10 .

Feedback(attempt): The opposite of -10 is 10 and $|-10| = 10$, so they are the same.

Problem 12 Two numbers, a and b are opposites. If the distance between a and b on a number line is 10 units, what are the two numbers?

Multiple Choice:

(a) -10 and 10

(b) -5 and 5 ✓

(c) -6 and 4

(d) -7 and 3

Hint: Draw a number line and with a and b ten units apart.

Feedback(correct): Good job! You made it through this problem and this section.